WORKED SOLUTION

STEP II 1997, Question 1

Pure • Digit Sums and Counting

> **STEP II 1997, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/97-s2-q1

The key idea

Don't list the numbers: count them, then add up the contribution of each place value separately. Every allowed digit turns up equally often in each of the last three places, so the sums practically do themselves. Better still, there is a symmetry: replacing each digit d by $7-d$ swaps $0 \leftrightarrow 7$ and $2 \leftrightarrow 5$, so the numbers pair off with a common sum, and the factorisation drops straight out.

Solution

Which numbers qualify?

A qualifying number between 1000 and 6000 has four digits, and its thousands digit must be one of $0, 2, 5, 7$. It can't be 0 (that gives fewer than four digits) and it can't be 7 (that gives a number over 7000). So the thousands digit is 2 or 5, and every number from 2000 to 5777 built this way lies in range. The endpoints 1000 and 6000 themselves contain the digits 1 and 6, so it doesn't matter whether "between" includes them.

The other three digits can each be any of the four allowed digits, independently. So there are

$$2 \times 4 \times 4 \times 4 = 128$$

numbers.

Adding by place value

Thousands. Half of the 128 numbers, that is 64, start with 2 and the other 64 start with 5. The thousands digits contribute

$$64 \times (2+5) \times 1000 = 448\,000.$$

Hundreds, tens and units. Fix one of these places. Each of the four digits appears there in exactly $128/4 = 32$ of the numbers, because the remaining digits can be chosen in the

same number of ways whichever digit sits in that place. The digit total in that place is therefore $32 \times (0+2+5+7) = 32 \times 14 = 448$. The three places carry weights 100, 10 and 1, so together they contribute

$$448 \times (100+10+1) = 448 \times 111 = 49\,728.$$

Adding, the required sum is

$$S = 448 \times 1000 + 448 \times 111 = 448 \times 1111 = 497\,728.$$

As a product of primes

We have $448 = 64 \times 7 = 2^6 \times 7$, and $1111 = 11 \times 101$, where 101 is prime (it isn't divisible by 2, 3, 5 or 7, and $11^2 > 101$). Hence

$$S = 2^6 \times 7 \times 11 \times 101.$$

Ask yourself Is there a way to see the answer without any place-value bookkeeping?

Yes. If n qualifies, so does $7777 - n$: subtracting from 7777 digit by digit involves no borrowing, and it sends each digit d to $7 - d$, which maps $\{0,2,5,7\}$ to itself and swaps the leading digits 2 and 5. So the 128 numbers split into 64 pairs, each adding to 7777 (no number is paired with itself, since 7777 is odd). Therefore

$$S = 64 \times 7777 = 2^6 \times 7 \times 1111 = 2^6 \times 7 \times 11 \times 101,$$

in agreement with the work above.

Checks

A short computer search over 1000 to 6000 finds exactly 128 qualifying numbers with sum 497 728, and $2^6 \times 7 \times 11 \times 101 = 64 \times 7777 = 497\,728$. As a sanity check on size, the average qualifying number is $7777/2 = 3888.5$, comfortably in the middle of the range 2000 to 5777.

Common slips

- Allowing 7 as the leading digit (giving numbers over 6000) or forgetting that 0 cannot lead.
- Saying each digit appears $128/4$ times in the thousands place too: only 2 and 5 appear there, 64 times each.
- Stopping at 497 728, or at 448×1111, without breaking 1111 into 11×101.

Beyond STEP

Counting how often each digit appears in each place is the simplest instance of a *double-counting* argument: summing the same quantity in two orders, here by number and by place. The $7777 - n$ pairing is a symmetry argument of the kind that also gives Gauss's formula $1 + 2 + \cdots + n = \frac{1}{2}n(n+1)$. Numbers whose digits come from a fixed set, and their sums, are a staple of elementary number theory and combinatorics; Davenport's *The Higher Arithmetic* is a lovely next step.

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